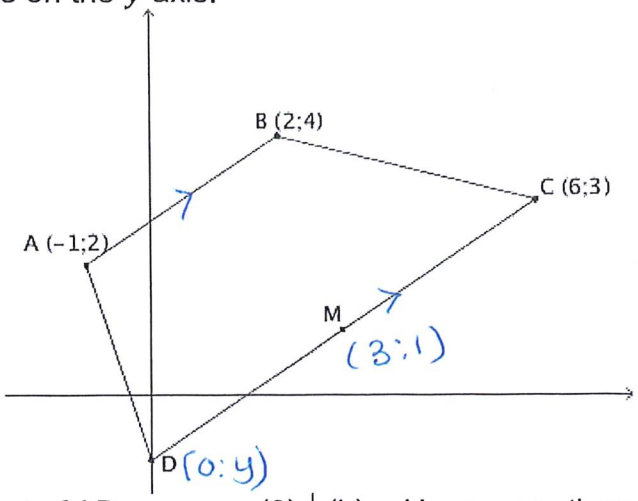
Page a day worksheet **PAD nr: 15**

Grade 10 – Analytical Geometry

Name:	Memo

In the following diagram we have A (-1;2), B(2;4), C(6;3) and D $(x; y)$ forming a trapezium with AB||CD and D a coordinate on the y-axis.



(a) Determine the gradient of AB. (2)

$m_{AB} = \frac{4-2}{2--1} = \frac{2}{3}$

(b) Hence, or otherwise, determine the coordinates of D. (3)

$m_{DC} = m_{AB}$ (AB // DC)

$\frac{3-y}{6-0} = \frac{2}{3}$

$3-y = 4$

$-1 = y$ D (0; -1)

(c) If M is the midpoint of CD and D's y-coordinate -1, determine M's coordinates. (2)

$M(x;y) = \left(\frac{0+6}{2}; \frac{-1+3}{2}\right)$

M (3; 1)

(d) Is ABCD an isosceles trapezium? Show all calculations to justify your statement. (4)

$AD = \sqrt{(-1-0)^2+(2--1)^2} = \sqrt{10}$

$BC = \sqrt{(2-6)^2+(4-3)^2} = \sqrt{17}$

$AD \neq BC$ ∴ not isos trapezium

(e) Determine the equation of AB. (2)

$y = \frac{2}{3}x + c$ Sub B

$4 = \frac{2}{3}(2) + c$

$\frac{8}{3} = c$

$y = \frac{2}{3}x + \frac{8}{3}$

(f) Is $AB \perp CD$? Show all calculations. (3)

No AB // CD (given)

$m_{AB} = m_{CD}$

$m_{AB} \times m_{CD} = \frac{2}{3} \times \frac{2}{3} = \frac{4}{9}$

$m_{AB} \times m_{CD} \neq -1$

∴ not ⊥

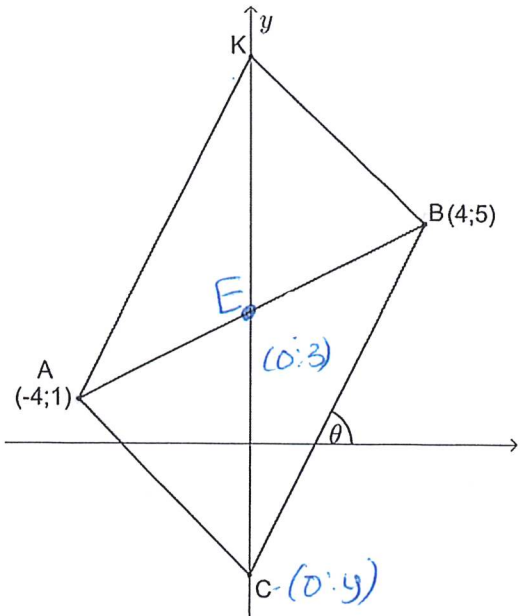
[16]

Page a day worksheet **PAD nr: 16**

Grade 10 – Analytical Geometry

Name:	Memo

The diagram below shows quadrilateral AKBC, with A (-4;1) and B (4;5). C is on the y-axis and K is the fourth vertex of the quadrilateral. The gradient of the line segment BC is 2.



(a) Determine the coordinates of the midpoint of AB. (2)

$(x;y) = \left(\frac{-4+4}{2}; \frac{1+5}{2}\right)$

$E = (0; 3)$

(b) Show that the coordinates of C are (0;-3). (3)

$m_{BC} = \frac{5-y}{4-0} = \frac{2}{1}$ (Given $m=2$)

$5-y = 8$

$-3 = y$ $C(0;-3)$

(c) Determine K's coordinates such that AKBC is a parallelogram. (2)

$C \rightarrow B \rightarrow (x+4; y+8)$

$\therefore A \rightarrow K \rightarrow (-4+4; 1+8)$

$K(0;9)$

(d) Determine the equation of AB. (4)

$m = \frac{5-1}{4-(-4)} = \frac{4}{8} = \frac{1}{2}$

$y = \frac{1}{2}x + c$ sub $B(4;5)$

$5 = \frac{1}{2}(4) + c$

$3 = c$ $y = \frac{1}{2}x + 3$

(e) Will A, B and M (8;9) be collinear? Show all calculations to justify your answer. (4)

$m_{AB} = \frac{1}{2}$ $m_{MB} = \frac{9-5}{8-4} = \frac{4}{4} = 1$

$m_{AB} \neq m_{MB}$

$\therefore$ not collinear

(f) Is AC⊥BC? Show all calculations. (3)

$m_{BC} = \frac{5-(-3)}{4-0} = \frac{8}{4} = 2$

$m_{AC} = \frac{1-(-3)}{-4-0} = \frac{4}{-4} = -1$

$m_{AC} \times m_{BC}$

$= 2 \times -1$

$= -2$.

$\therefore$ Not ⊥

[16]